

Curriculum for Undergraduate Degree (B.Tech.) in Computer Science and Engineering (w.e.f. AY: 2020-21)

Part III: Detailed Curriculum

Seventh Semester

Course Name:	Quantum Computing		
Course Code:	PE-CS701A	Category:	Professional Elective Courses
Semester:	7th	Credit:	3.0
L-T-P:	3-0-0	Pre-Requisites:	Basic understanding of Quantum Theory, Linear Algebra, Theory of Computation
Full Marks:	100		
Examination Scheme:	Semester Examination:70	Continuous Assessment:25	Attendance:05

Course Objectives:

1	Inculcate the idea of Quantum Computing and its inevitability in the near future.
2.	Develop a basic idea on Qubit, Quantum gates, Quantum Cryptography and appreciate the power of parallel computing with few quantum algorithms.
3.	Get hands-on experience in building quantum circuits using IBM's Qiskit.

Course Contents:

Module No.	Description of Topic	Contact Hrs.
1.	Vector and Operators: ✓ The Motivation: Shannon's Information theory & its connection to Entropy ✓ Concept of Vector Spaces. ✓ Basis & dimensions ✓ Linear Combination of Vectors: Representing a quantum state by a vector ✓ Uniqueness of a spanning set, Inner Product (its relation to the vector dot product), Orthonormality, Gram-Schmidt orthogonalization, Bra-ket formalism and its usefulness, the Cauchy schwarz Inequality. ✓ Outer Products, The Closure Relation ✓ Concept of Operators in Quantum Mechanics and their representation using matrices, Hermitian & Unitary operators ✓ Concept of Eigen values & Eigen Vectors and the span of vector space, Spectral Decomposition, Trace of an operator, important properties of Trace, Expectation Value of Operator, Projection Operator, Positive Operators	10
	Tensor Products:	

2.	✓ Representing Composite States in Quantum Mechanics, Tensor products of column vectors, operators and tensor products of matrices ✓ Concept of Pure and Mixed State ✓ Density Operator: Its usefulness ✓ Density Operator of Pure & Mixed state, Key Properties, Characterizing Mixed State	7
3.	The Building Block: ✓ Concept of Qubit: its difference with a conventional Bit. Bloch vector & Qubit. ✓ Pauli spin matrices, Quantum Gates: Pauli matrices as Phase & NOT gates. ✓ Irreversibility of classical gates, Introduction to Hadamard, CNOT and Toffoli gates with their truth tables. ✓ Entangled states: Its concept and mechanism of creation using quantum gates. Quantum teleportation using entanglement.	8
4.	Parallel computing: ✓ Deutsch Jozsa algorithm ✓ Unstructured search using Grover's algorithm. ✓ Quantum key distribution: The future of Cryptography	6
5.	Hands-on: ✓ Recent trends in Quantum Computing ✓ Learning to use IBM's Qiskit to build quantum circuits	5
Total		36L

Course Outcomes:

After completion of the course, students will be able to:

1.	Relate vectors to physical states of system and matrices to operators.
2.	Examine the application of various quantum gates on qubit.
3.	Discover the power of parallel computing using quantum algorithms.
4.	Construct simple quantum circuits by IBM's Qiskit.

Learning Resources:

1.	'Quantum Computing without Magic' by Zdzislaw Meglicki
2.	'Quantum Computing Explained' by David Mc Mahon
3.	'Quantum Computer Science' by Marco Lanzagorta and Jeffrey Uhlmann
4.	'An Introduction to Quantum Computing' by Phillip Kaye, Raymond Laflamme and Michele Mosca.
5.	https://qiskit.org/textbook/preface.html

Course Name:	Cloud Computing		
Course Code:	PE-CS701B	Category:	Professional Elective Courses
Semester:	Seventh	Credit:	3
L-T-P:	3-0-0	Pre-Requisites:	Computer Architecture, Operating System
Full Marks:	100		
Examination Scheme:	Semester Examination:70	Continuous Assessment:25	Attendance:05

Course Objectives:

1	This course gives students an insight into the basics of cloud computing along with virtualization.
2	It will provide the students basic understanding of cloud security and privacy issues.
3	Students will be able to use different cloud services for different purposes.

Course Contents:

Module No.	Description of Topic	Contact Hrs.
1	<u>Definition of Cloud Computing and its Basics:</u> Defining a Cloud, Cloud Types – NIST model, Cloud Cube model, Deployment models (Public, Private, Hybrid and Community Clouds), Service model: - Infrastructure as a Service, Platform as a Service, Software as a Service with examples of services/ service providers. Cloud Reference model, Characteristics of Cloud Computing, Benefits and advantages of Cloud Computing, A brief introduction on Composability, Infrastructure, Platforms, Virtual Appliances, Communication Protocols, Applications, Connecting to the Cloud by Clients, IaaS –concept, Workload, partitioning of virtual private server instances, Pods, aggregations. SaaS - Basic concept and characteristics, Open SaaS and SOA, examples of SaaS platform Identity as a Service (IDaaS) Compliance as a Service (CaaS)	9
2	<u>Use of Platforms in Cloud Computing:</u> Concepts of Abstraction and Virtualization technologies: Types of virtualization (access, application, CPU, storage), Load Balancing and Virtualization: Basic Concepts, Network resources for load balancing, Advanced load balancing (including Application Delivery Controller and Application Delivery Network), Mention of The Google Cloud as an example of use of load balancing. Hypervisors: Virtual machine technology and types, VMware vSphere Machine Imaging. Porting of applications in the Cloud: The simple Cloud API and AppZero Virtual Application appliance, Concepts of Platform as a Service, Definition of services, Distinction between SaaS and PaaS (knowledge of Salesforce.com and Force.com), Application development Use of PaaS Application frameworks. Discussion of Google Applications Portfolio – Indexed search, Dark Web, Aggregation and disintermediation, Productivity applications and service, Adwords, Google Analytics, Google Translate, a brief discussion on Google	12

	Toolkit (including introduction of Google APIs in brief), major features of Google App Engine service. Windows Azure platform: Microsoft's approach, architecture, and main elements, overview of Windows Azure AppFabric, Content Delivery Network, SQL Azure, and Windows Live services.	
3	<u>Cloud Infrastructure:</u> Cloud Management: An overview of the features of network management systems and a brief introduction of related products from large cloud vendors, Monitoring of an entire cloud computing deployment stack – an overview with mention of some products, Lifecycle management of cloud services (six stages of lifecycle). Concepts of Cloud Security: Cloud security concerns, Security boundary, Security service boundary Overview of security mapping Security of data: Brokered cloud storage access, Storage location and tenancy, encryption, and auditing and compliance Identity management (awareness of Identity protocol standards).	7
4	<u>Concepts of Services and Applications :</u> Service Oriented Architecture: Basic concepts of message-based transactions, Protocol stack for an SOA architecture, Event-driven SOA, Enterprise Service Bus, Service catalogs, Applications in the Cloud: Concepts of cloud transactions, functionality mapping, Application attributes, Cloud service attributes, System abstraction and Cloud Bursting, Applications and Cloud APIs Cloud-based Storage: Cloud storage definition – Manned and Unmanned Webmail Services: Cloud mail services including Google Gmail, Mail2Web, Windows Live Hotmail, Yahoo mail, concepts of Syndication services.	8
Total		36L

Course Outcomes:

After completion of the course, students will be able to:

1	Explain the fundamental concept of cloud computing and its characteristics, benefits and limitations.
2	Classify different types of cloud models, architecture and infrastructure of cloud computing and its examples.
3	Demonstrate abstraction and different types of virtualization, load balancing technology and their role in the cloud computing model.
4	Solve the security, privacy issues and cloud management of cloud computing.
5	Use various cloud services in different applications.

Learning Resources:

1	Cloud Computing Bible by Barrie Sosinsky, Wiley India Pvt. Ltd, 2013
2	Mastering Cloud Computing by Rajkumar Buyya, Christian Vecchiola, S. Thamarai Selvi, McGraw Hill Education (India) Private Limited, 2013
3	Cloud computing: A practical approach, Anthony T. Velte, Tata Mcgraw-Hill
4	Cloud Computing, Miller, Pearson
5	Building applications in cloud: Concept, Patterns and Projects, Moyer, Pearson
6	Cloud Computing – Second Edition by Dr. Kumar Saurabh, Wiley India

Course Name:	Neural Networks and Deep Learning		
Course Code:	PE-CS701C	Category:	PE
Semester:	7th	Credit:	3
L-T-P:	3-0-0	Pre-Requisites:	BS-M101, BS-M301
Full Marks:	100		
Examination Scheme:	Semester Examination:70	Continuous Assessment:25	Attendance:05

Course Objectives:	
1	To explore the evolution of ANN starting from its initial phase
2	To learn the structure and function of ANN
3	To analyse ANN learning
4	To explore different deep neural networks
5	To learn tools, applications, limitations and future scopes

Course Contents:		
Module No.	Description of Topic	Contact Hrs.
1	Introduction: McCulloch-Pitts Neuron, Perceptron, Perceptron Learning Algorithm and Pattern classification using perceptron, Perceptron function vs Sigmoid or Logistic function, Sigmoid Neuron.	4
2	Structural and Functional framework of ANN: Feedforward Neural Networks: Activation Functions-Hidden Layer, Activation Functions-Output Layer, Multilayer feedforward neural networks with example.	4
3	Learning: Approximation of any arbitrary function, Error or Loss Function- Mean Square and Cross Entropy, Learning Algorithm-Minimization of Loss, Gradient Descent, Backpropagation of error with example, Optimizers, Learning Rate, Overfitting and Underfitting, L1 and L2 Regularization, Dropout, Early Stopping, Augmentation.	8
4	Deep Neural Networks: Convolutional Neural Network (CNN), Pre-Trained Networks, Recurrent Neural Network (RNN) - LSTM, Restricted Boltzmann Machine (RBM), Deep Belief Network (DBN), Autoencoders, Diffusion Model.	8
5	Generative Adversarial Net (GAN) - Introduction, Applications of GANs, GAN Discriminator, GAN Generator, GAN Training - Upsampling, Transposed Convolutions, Binary Cross Entropy (BCE) Loss for GANs.	6
6	Deep Learning Tools, Research Applications, Limitations of Deep Learning and Potential Future Directions.	6
Total		36L

Course Outcomes:	
After completion of the course, students will be able to:	
1	Explain the concepts of McCulloch-Pitts Neuron, Perceptron and Sigmoid Neuron
2	Describe the structural and functional framework of Neural Networks
3	Interpret the learning procedure of Neural Networks
4	Classify different deep learning models
5	Recognize different deep learning tools, the research applications, limitations and future directions

Learning Resources:	
1	Neural Networks and Learning Machines, Simon Haykin, Pearson; 3rd edition, 2009.
2	Goodfellow, I., Bengio, Y., and Courville, A., Deep Learning, MIT Press, 2016.
3	Deep Learning: A Practitioner's Approach, Josh Patterson, Adam Gibson, O'Reilly, 2017
4	Fundamentals of Deep Learning, Nithin Buduma, Nikhil Buduma, Joe PapaO'Reilly Media; 2nd edition, 2022.
	Online Resources:
1	https://www.deeplearningbook.org/
2	https://d2l.ai/
3	Modeling Tools: https://www.tensorflow.org/ , https://pytorch.org/ , http://caffe.berkeleyvision.org/ , https://theano-

Course Name:	Natural Language Processing		
Course Code:	PE- CS702A	Category:	Professional Elective Courses
Semester:	7 th	Credit:	3
L-T-P:	3-0-0	Pre-Requisites:	Formal Language and Automata Theory
Full Marks:	100		
Examination Scheme:	Semester Examination: 70	Continuous Assessment: 25	Attendance: 05

Course Objectives:	
1	To familiarize the concepts and techniques of Natural language Processing
2	To introduce basic mathematical models and methods used in NLP applications to formulate computational solutions
3	To provide students with the knowledge on designing procedures for natural language resource annotation and the use of related tools for text analysis using such tools
4	To introduce students to research and development work in information retrieval, information extraction, and knowledge discovery using different natural language resources

Course Contents:		
Module No.	Description of Topic	Contact Hrs.
1	Introduction: Human Languages, Phases in natural language processing, Review of Regular Expressions, Finite State Automata, CFG and Different Parsing techniques.	3
2	Tokenization: Word tokenization, Normalization, Sentence segmentation	2
3	Morphology: Inflectional and Derivational Morphology, Finite State Morphological Parsing, The Lexicon and Morphotactic, Morphological Parsing with Finite State Transducers, Orthographic Rules and Finite State Transducers, Porter Stemmer	5
4	Language Models: Introduction to N-grams, Chain Rule, Smoothing–Add-One Smoothing, Witten-Bell Discounting; Back off, Deleted Interpolation, N-grams for Spelling and Word Prediction, Evaluation of language models.	6
5	Text Classification: Text Classification, Naïve Bayes’ Text Classification, Evaluation, Sentiment Analysis – Opinion Mining and Emotion Analysis, Resources and Techniques.	5
6	POS Tagging: Tagsets, HMM Part-of-Speech Tagging, Markov Chain, Hidden Markov Models, Viterbi Algorithm, Rule based and Machine Learning based approaches, Evaluation	8
7	Computational Lexical Semantics: Introduction to Lexical Semantics – Homonymy, Polysemy, Synonymy, Thesaurus – WordNet, Computational Lexical Semantics – Thesaurus-based and Distributional Word Similarity	3
8	Information Retrieval: Boolean Retrieval, Term document incidence, The Inverted Index, Query Optimization, Phrase Queries, Ranked Retrieval – Term Frequency – Inverse Document Frequency based ranking, Zone Indexing, Query term proximity, Cosine ranking,	4

	Combining different features for ranking, Search Engine Evaluation, Relevance Feedback	
Total		36L

Course Outcomes:

After completion of the course, students will be able to:

1	Identify the basic concepts of Natural language processing.
2	Describe the concepts of morphology, Syntactic processing, and semantics analysis of natural language processing.
3	Illustrate different language Models and Text classification techniques
4	Use appropriate statistical models for a given natural language problem
5	Use information retrieval techniques to solve real life problems.

Learning Resources:

1	Speech and Language Processing, Jurafsky and Martin, Pearson Education
2	Foundation of Statistical Natural Language Processing, Manning and Schutze, MIT Press
3	Multilingual Natural Language Processing Applications from Theory to Practice: Bikel, Pearson.

Course Name:	Block Chain Technology		
Course Code:	PE-CS702B	Category:	Professional Elective Course
Semester:	Seventh	Credit:	3
L-T-P:	3-0-0	Pre-Requisites:	Basic concepts of Computer Networks and Operating Systems
Full Marks:	100		
Examination Scheme:	Semester Examination: 70	Continuous Assessment: 25	Attendance: 05

Course Objectives:	
1	To understand the distributed decentralized database
2	To familiarize with fundamentals of Blockchain and its various applications
3	To identify distributed ledger technologies and their architecture.
4	To describe the Hashing in Blockchain mining
5	To get acquaintance with the Ethereum Virtual Machine (EVM), clients of EVM, and Ethereum Key pairs
6	To know about the cryptography and Bitcoin

Course Contents:		
Module No.	Description of Topic	Contact Hrs.
1	Basics of Blockchain: Introduction, Concept of Blockchain, History, Definition, Fundamentals & Characteristics of Blockchain, Public, Private and Hybrid Blockchains, Distributed Ledger Technologies, Architecture of Blockchain, Transactions, Chaining Blocks. Introduction of Decentralized System, Distributed Decentralized Databases, Decentralization.	6L
2	Hash Functions: Introduction, Hashing, Message Authentication Code, Secure Hash algorithm, Distributed Hash Tables, Hashing and Data Structures, Hashing in Blockchain Mining. Consensus: Introduction, Consensus Approach, Consensus Algorithms, Byzantine Agreement Methods.	6L
3	Blockchain Components: Introduction, Ethereum, History, Ethereum Virtual Machine, Working of Ethereum, Ethereum Clients, Ethereum Key Pairs, Ethereum Addresses, Ethereum Wallets, Ethereum Transactions, Ethereum Languages, Ethereum Development Tools.	6L
4	Cryptography: Introduction, Cryptography primitives, Symmetric Cryptography, Asymmetric Cryptography. Smart Contracts: Introduction, Absolute and Immutable, Contractual Confidentiality, Characteristics, Use cases.	6L
5	Bitcoins: Introduction, Working of Bitcoin, Creation of Bitcoins, Markle Trees, Bitcoin Block Structure, Bitcoin Address, Bitcoin Transactions, Bitcoin Network, Bitcoin Wallets, Bitcoin Payments, Bitcoin Clients, Bitcoin Supply.	6L

6	Decentralized Applications: Introduction, Today's Web applications Requirement, Mining in Blockchain, Blockchain in healthcare, safety and security, Validation and Identification of Bitcoin based task, Mining Hardware and Software, Bitcoins Management.	6L
Total		36L

Course Outcomes:		
After completion of the course, students will be able to:		
1	Describe the basic concepts and working of Blockchain Technology including Hash function and Consensus algorithms	
2	Explain the design principles of Bitcoin, Ethereum, Smart Contracts and role of cryptography in these architectures.	
3	Explain the working of smart contracts in developing different Decentralized applications with knowledge of Bitcoin concept.	

Learning Resources:	
1	Mastering Blockchain: A deep dive into distributed ledgers, consensus protocols, smart contracts, DApps, cryptocurrencies, Ethereum, and more, 3rd Edition, Imran Bashir, Packt Publishing, 2020, ISBN: 9781839213199, book website: https://www.packtpub.com/product/mastering-blockchain-third-edition/9781839213199
2	Blockchain Technology-Concepts and Applications by Kumar Saurabh and Ashutosh Saxena, Wiley Publishers.
3	Josh Thompson, 'Blockchain: The Blockchain for Beginnings, Guild to Blockchain Technology and Blockchain Programming', Create Space Independent Publishing Platform, 2017.
4	Hyperledger Tutorials - https://www.hyperledger.org/use/tutorials
5	Ethereum Development Resources - https://ethereum.org/en/developers

Course Name:	Artificial Intelligence		
Course Code:	PE-CS702C	Category:	Professional Elective
Semester:	7th	Credit:	3
L-T-P:	3-0-0	Pre-Requisites:	Discrete Mathematics, Algorithm
Full Marks:	100		
Examination Scheme:	Semester Examination: -70	Continuous Assessment: - 25	Attendance: -5

Course Objectives:	
1	This subject will help in acquiring knowledge on Basic Techniques of AI and Intelligent agents.
2	The subject will help to acquire knowledge on reasoning with and without uncertainty.
3	The subject will help in having elementary knowledge on expert system development in various domain.

Course Contents:		
Module No.	Description of Topic	Contact Hrs.
1	Introduction Overview of Artificial intelligence- Problems of AI, AI technique, Tic - Tac - Toe problem. Intelligent Agents Agents & environment, nature of environment, structure of agents, goal based agents, utility-based agents, learning agents. Problem Solving Problems, Problem Space & search: Defining the problem as state space search, production system, problem characteristics, issues in the design of search programs.	8
2	Search techniques Solving problems by searching: problem solving agents, searching for solutions; uniform search strategies: breadth first search, depth first search, depth limited search, bidirectional search, comparing uniform search strategies. Heuristic search strategies Greedy best-first search, A* search, memory bounded heuristic search: local search algorithms & optimization problems: Hill climbing search, simulated annealing search, local beam search, genetic algorithms; constraint satisfaction problems, local search for constraint satisfaction problems. Adversarial search Games, optimal decisions & strategies in games, the minimax search procedure, alpha-beta pruning, additional refinements, iterative deepening.	15
3	Using predicate logic Representing simple fact in logic, representing instant & ISA relationship, computable functions & predicates, resolution, natural deduction. Probabilistic reasoning Representing knowledge in an uncertain domain, the semantics of	6

	Bayesian networks, Dempster-Shafer theory, Fuzzy sets & fuzzy logics.	
4	Applications of AI in the following domain (a) E-Commerce (b) Expert Education (c) Finance (d) Social media (e) Robotics. Preliminary knowledge of Chatbots and AI based Games .	7
Total		36L

Course Outcomes:

After completion of the course, students will be able to:

1.	Identify the basic concepts of AI Techniques, Agents, and Production System
2.	Compare between heuristic, non-heuristic search strategies and adversarial search
3.	Apply Predicate logic, Resolution and Probability based inference in different problems
4.	Explain the expert system architecture in different domains

Learning Resources:

1.	Artificial Intelligence, Ritch & Knight, TMH
2.	Artificial Intelligence A Modern Approach, Stuart Russel Peter Norvig Pearson
3.	Introduction to Artificial Intelligence & Expert Systems, Patterson, PHI
4.	Poole, Computational Intelligence, OUP
5.	Expert Systems, Giarranto, VIKAS
6.	M.C. Trivedi, Artificial Intelligence, Khanna Publishing House, New Delhi (AICTE Recommended Textbook – 2018)

Open Elective -I

Subject Code	Subject Name	Offered by
OE-EC701A	Adhoc and Sensor Network	ECE Department
OE-EE701B	Control Systems	EE Department
OE-IT701D	Multimedia Systems	IT Department
OE-M701A	Operations Research and Optimizing Technique	BS Department
OE-ME701B	Mechatronic Systems	ME Department

Course Name:	Adhoc and Sensor Network		
Course Code:	OE-EC701A	Category:	Open Elective Course
Semester:	Seventh	Credit:	3
L-T-P:	3-0-0	Pre-Requisites:	Computer Networks
Full Marks:	100		
Examination Scheme:	Semester Examination: 70	Continuous Assessment: 25	Attendance: 05

Course Objectives:	
1	Learn Ad hoc network and Sensor Network fundamentals
2	Understand the different routing protocols
3	Have an in-depth knowledge on sensor network architecture and design issues
4	Understand the transport layer and security issues possible in Ad hoc and Sensor networks
5	Have an exposure to mote programming platforms and tools

Course Contents:		
Module No.	Description of Topic	Contact Hrs.
1	Adhoc Networks – Introduction & Routing Protocols : Elements of Ad hoc Wireless Networks, Issues in Ad hoc wireless networks, Example commercial applications of Ad hoc networking, Ad hoc wireless Internet, Issues in Designing a Routing Protocol for Ad Hoc Wireless Networks, Classifications of Routing Protocols, Table Driven Routing Protocols – Destination Sequenced Distance Vector (DSDV), On-Demand Routing protocols –Ad hoc On-Demand Distance Vector Routing (AODV).	8
2	Sensor Networks – Introduction & Architecture : Challenges for Wireless Sensor Networks, Enabling Technologies for Wireless Sensor Networks, WSN application examples, Single-Node Architecture – Hardware Components, Energy Consumption of Sensor Nodes, Network Architecture – Sensor Network Scenarios, Transceiver Design Considerations, Optimization Goals and Figures of Merit.	8

3	WSN Networking Concepts & Protocols : MAC Protocols for Wireless Sensor Networks, Low Duty Cycle Protocols And Wakeup Concepts – S-MAC, The Mediation Device Protocol, Contention based protocols – PAMAS, Schedule based protocols – LEACH, IEEE 802.15.4 MAC protocol, Routing Protocols- Energy Efficient Routing, Challenges and Issues in Transport layer protocol.	8
4	Sensor Network Security : Security Requirements, Issues and Challenges in Security Provisioning, Network Security Attacks, Layer wise attacks in wireless sensor networks, possible solutions for jamming, tampering, black hole attack, flooding attack. Key Distribution and Management, Secure Routing – SPINS, reliability requirements in sensor networks.	6
5	Sensor Network Platforms and Tools : Sensor Node Hardware – Berkeley Motes, Programming Challenges, Node-level software platforms – TinyOS, nesC, CONTIKIOS, Node-level Simulators – NS2 and its extension to sensor networks, COOJA, TOSSIM, Programming beyond individual nodes – State centric programming	6
Total		36

Course Outcomes:

After completion of the course, students will be able to:

1	Know the basics of Ad hoc networks and Wireless Sensor Networks
2	Apply this knowledge to identify the suitable routing algorithm based on the network and user requirement
3	Apply the knowledge to identify appropriate physical and MAC layer protocols.
4	Explain the transport layer and security issues possible in Ad hoc and sensor networks.
5	Familiar with the OS used in Wireless Sensor Networks and build basic modules.

Learning Resources:

1	C. Siva Ram Murthy and B. S. Manoj, “Ad Hoc Wireless Networks Architectures and Protocols”, Prentice Hall, PTR, 2004.
2	Kazem Sohraby, Daniel Minoli, & Taieb Znati, “Wireless Sensor Networks-Technology, Protocols, and Applications”, John Wiley, 2007.
3	Carlos De Moraes Cordeiro, Dharma Prakash Agrawal “Ad Hoc & Sensor Networks: Theory and Applications”, World Scientific Publishing Company, 2006.
4	Holger Karl, Andreas Willig “Protocols and Architecture for Wireless Sensor Networks” John Wiley and Sons, Ltd.

Course Name:	Control System		
Course Code:	OE-EE701B	Category:	Open Elective Course
Semester:	7 th	Credit:	3
L-T-P:	3-0-0	Pre-Requisites:	1.Basic Electrical Engineering 2.Electric Circuit Theory 3.Electric Machine
Full Marks:	100		
Examination Scheme:	Semester Examination: 70	Continuous Assessment: 25	Attendance: 05

Course Objectives:	
1	To classify open loop and closed loop control system and the effect of feedback system
2	To describe transfer function of linear dynamic system using linear differential equations
3	To compare time response and frequency response analysis of linear system.
4	To understand stability analysis of different systems.
5	To explain the improvement of system performance using compensator or controller
6	To develop state space model formation of some linear system.

Course Contents:		
Module No.	Description of Topic	Contact Hrs.
1	Introduction to control system: Objectives of control system, Concept of feedback and Effects of feedback examples of feedback control systems. Types of control systems, Definition of linear time varying, linear time invariant (LTI) and nonlinear systems, Transfer function concept. Pole and Zeroes of a transfer function. Properties of Transfer function.	3L
2	Mathematical modeling of dynamic systems: Translational systems, Rotational systems, Electrical analogy of Spring–Mass-Dashpot system. Block diagram representation of control systems. Block diagram algebra. Signal flow graph. Mason's gain formula. Block diagram level description of feedback control systems for position control, speed control of DC motors.	7L
3	Time domain analysis: Time domain analysis of first and standard second order closed loop system as step and impulse input. Concept of undamped natural frequency, damping, overshoot, rise time and settling time. Dependence of time domain performance parameters on natural frequency and damping ratio. Effects of Pole and Zeros on transient response. Stability by pole location. Routh-Hurwitz criteria and applications. Concepts of system types and error constants. Steady state errors analysis in control systems due to step, ramp and parabolic inputs.	7L

4	Stability Analysis: Root locus techniques, construction of Root Loci for simple linear time invariant systems. Effects of gain on the movement of Pole and Zeros. Frequency domain analysis of linear system: Bode plots, Polar plots, Concept of resonance frequency of peak magnification. Nyquist criteria, measure of relative stability, phase and gain margin. Determination of margins in Bode plot.	10L
5	Compensator and controller: Improvement of system performance through compensation. Lead, Lag and Lead- lag compensation, and P, PI, PD and PID control.	5L
6	State variable Analysis: Concepts of state variables. State space model formation of physical system. Solution of state equations. Eigen values, State transition matrix and its properties. Concept of controllability and observability.	5L
7	Nonlinear control systems: Basic concepts and stability analysis, Describing function, Phase plane method.	3L
Total		40L

Course Outcomes:

After completion of the course, students will be able to:

1	Develop mathematical model and compute transfer of linear system.
2	Calculate peak time, rise time, settling time, steady state error of linear system in time response analysis.
3	Compute peak response, bandwidth, gain crossover, phase crossover, gain margin and phase margin of linear system in frequency response analysis.
4	Explain stability analysis of linear system in time domain and frequency domain approach.
5	Describe the effect of using controller and compensator to improve the system performances with given criterion.
6	Apply the knowledge of state variable techniques for analysis of multivariable linear systems.
7	Explain the basic concept and analysis of the nonlinear control system.

Learning Resources:

Recommended Text Books

1	Modern Control Engineering, K. Ogata, 4th Edition, Pearson Education
2	Control System Engineering, I. J. Nagrath & M. Gopal. New Age International Publication.

Alternative Text Books

3	Automatic Control Systems, B.C. Kuo & F. Golnaraghi, 8th Edition, PHI
4	Control System Engineering, D. Roy Choudhury, PHI

Reference Books

5	Control System Engineering, R. Anandanatarajan & R. Ramesh Babu, SCITECH
6	Control System Engineering, Norman Nise, 5th Edition, John Wiley & Sons
7	Control Engineering Theory & Practice, Bandyopadhyay, PH

Course Name:	Multimedia Systems		
Course Code:	OE-IT701D	Category:	Open Elective Courses
Semester:	7 th	Credit:	3
L-T-P:	3-0-0	Pre-Requisites:	Fundamental knowledge of Computation, Networking and DBMS
Full Marks:	100		
Examination Scheme:	Semester Examination: 70	Continuous Assessment: 25	Attendance: 05

Course Objectives:	
1	To give each student a firm grounding in the fundamentals of the underpinning technologies in graphics and multimedia.
2	To teach students the principled design of effective media for entertainment, communication, training and education.
3	To provide each student with experience in the generation of animations, virtual environments and multimedia applications, allowing the expression of creativity.

Course Contents:		
Module No.	Description of Topic	Contact Hrs.
1	Introduction: Multimedia today, Impact of Multimedia, Multimedia Systems, Components and Its Applications	2L
2	Text and, Image: Text: Types of Text, Ways to Present Text, Aspects of Text Design, Character, Character Set, Codes, Unicode, Encryption. Image: Formats, Image Color Scheme, Image Enhancement.	5L
3	Audio and Video: Audio: Basic Sound Concepts, Types of Sound, Digitizing Sound, Computer Representation of Sound (Sampling Rate, Sampling Size, And Quantization), Audio Formats, Audio tools, MIDI. Video: Analogue and Digital Video, Recording Formats and Standards (JPEG, MPEG, H.261) Transmission of Video Signals, Video Capture, and Computer based Animation.	6L
4	Synchronization, Storage models and Access Techniques: Temporal relationships, synchronization accuracy specification factors, quality of service, Magnetic media, optical media, file systems (traditional, multimedia) Multimedia devices – Output devices, CD-ROM, DVD, Scanner, CCD.	7L

5	Image and Video Database, Document Architecture and Content Management: Image representation, segmentation, similarity based retrieval, image retrieval by color, shape and texture; indexing- kd trees, R-trees, quad trees; Case studies- QBIC, Virage. Video Content, querying, video segmentation, indexing, Content Design and Development, General Design Principles Hypertext: Concept, Open Document Architecture (ODA), Multimedia and Hypermedia Coding Expert Group (MHEG), Standard Generalized Markup Language (SGML), Document Type Definition (DTD), Hypertext Markup Language (HTML) in Web Publishing. Case study of Applications.	13 L
6	Multimedia Applications: Interactive television, Video-on-demand, Video Conferencing, Educational Applications, Industrial Applications, Multimedia archives and digital libraries, media editors	3L
Total		36L

Course Outcomes:

After completion of the course, students will be able to:

1	Explain the concepts, principles and theories of Multimedia Applications and Virtual environments.
2	Demonstrate knowledge and understanding the various aspects and issues involved with development and deployment of multimedia system.
3	Analyze and solve problems related to their expertise in Multimedia Applications and Virtual Environments.
4	Demonstrate their ability to extend their basic knowledge to encompass new principles and practice.

Learning Resources:

1	"Multimedia: Computing, Communications & Applications" by Ralf Steinmetz and Klara Nahrstedt, Pearson Ed.
2	"Multimedia and Animation" by V.K. Jain, Khanna Publishing House, 2019.
3	"Multimedia Information System" by Nalin K. Sharda , PHI.
4	"Multimedia Communications" by Fred Halsall, Pearson Ed.
5	"Multimedia Systems" by Koegel Buford, Pearson Ed.
6	"Multimedia Literacy" by Fred Hoffstetter, McGraw Hill.
7	"Multimedia Fundamentals: Vol. 1- Media Coding and Content Processing" by Ralf Steinmetz and Klara Nahrstedt, PHI.
8	"Multimedia in Practice: Technology and Application" by J. Jeffcoate , PHI.

Course Name:	Operations Research and Optimizing Technique		
Course Code:	OE-M 701A	Category:	Optional Elective Courses
Semester:	Seventh	Credit:	3
L-T-P:	3-0-0	Pre-Requisites:	School mathematics, BS-M101, BS-M201
Full Marks:	100		
Examination Scheme:	Semester Examination: 70	Continuous Assessment: 25	Attendance: 05

Course Objectives:

1	To impart knowledge in concepts and tools of Operations Research
2	To understand mathematical models used in Operations Research
3	To apply these techniques constructively to make effective business decisions

Course Contents:

Module No.	Description of Topic	Contact Hrs.
1	Solving Linear Programming Problems : Formulation, Solving LPP: Using Simultaneous Equations and Graphical Method; Simplex, Duality, Big-M method, Transportation & Assignment, Travelling Salesman problem	11L
2	Game Theory : Introduction ; 2- person Zero – sum Game; Saddle Point; Mini-Max and Maxi-Min Theorems (statement only); Games without saddle point ; Graphical Method ; Principle of Dominance	7L
3	Queuing Theory : Introduction ; Basic Definitions and Notations ; Axiomatic Derivation of the 7L Arrival & Departure (Poisson Queue). Pure Birth and Death Models; Poisson Queue Models : M/M/1 : ∞ /FIFO and M/M/1: N/ FIFO.	6L
4	Network Analysis : Shortest Path : Floyd Algorithm ; Maximal Flow Problem (Ford-Fulkerson); PERT-CPM (Cost Analysis, Crashing, Resource Allocation excluded).	6L
5	Non-Linear Programming: Integer Programming, Dynamic Programming.	6L
Total		36L

Course Outcomes:

After completion of the course, students will be able to:	
1	Solve linear programming problems using appropriate techniques and optimization solvers, interpret the results obtained
2	Determine optimal strategy for Minimization of Cost of shipping of products from source to Destination/ Maximization of profits of shipping products using various methods, Finding initial basic feasible and optimal solution of the Transportation problems

3	Optimize the allocation of resources to Demand points in the best possible way using various techniques and minimize the cost or time of completion of number of jobs by number of persons
4	Extract the Optimum value in constrained and unconstrained situations
5	Acquire skills in analyzing queueing models.
6	Analyze competitive real-world phenomena using concepts from game theory. Analyze pure and mixed strategy games
7	Formulate Network models for service and manufacturing systems, and apply operations research techniques and algorithms to solve these Network problems

Learning Resources:

1	H. A. Taha, "Operations Research", Pearson
2	P. M. Karak – "Linear Programming and Theory of Games", ABS Publishing House
3	Ghosh and Chakraborty, "Linear Programming and Theory of Games", Central Book Agency
4	Ravindran, Philips and Solberg - "Operations Research", WILEY INDIA
5	Kanti Swaroop — "Operations Research", Sultan Chand & Sons
6	Rathindra P. Sen—"Operations Research: Algorithms and Applications", PHI
7	R. Panneerselvam - "Operations Research", PHI
8	A.M. Natarajan, P. Balasubramani and A. Tamilarasi - "Operations Research", Pearson
9	M. V. Durga Prasad – "Operations Research", CENGAGE Learning
10	J. K. Sharma - "Operations Research", Macmillan Publishing Company

Course Name:	Mechatronic Systems		
Course Code:	OE-ME701B	Category:	Open Elective Courses
Semester:	Seventh	Credit:	3
L-T-P:	3-0-0	Pre-Requisites:	Fluid Mechanics and Fluid Machinery, Kinematics and Theory of Machines, Basic Electrical Engineering, Basic Electronics Engineering
Full Marks:	100		
Examination Scheme:	Semester Examination: 70	Continuous Assessment: 25	Attendance: 05

Course Objectives:	
1	To make familiar about control system and power electronics in designing mechatronic system
2	To provide knowledge on electrical circuits, signal conditioning.

Course Contents:		
Module No.	Description of Topic/ Experiment	Contact Hrs.
1	Module 1: Introduction to Mechatronics: Definition, Mechatronics in design and manufacturing, Comparison between Traditional and Mechatronic approach; Concurrent engineering	3L
2	Module 2: Review of fundamentals of electronics: Logic gates and their operations, Signal processing devices, Data conversion devices, Input and output devices. Sensors and Transducers, Actuators, Limit switches, Relays	6L
3	Module 3: Control Systems: Open loop and closed loop control, block diagrams, transfer functions, Laplace transforms.	3L
4	Module 4: Electrical Drives: Stepper motors, servo drives.	2L
5	Module 5: Mechanical Drives: Different mechanisms, Ball screws, Linear motion bearings, Transfer systems.	3L
6	Module 6: Pneumatic and Hydraulic drives: elements of Pneumatic and Hydraulic drives, comparison between them. Design of pneumatic and hydraulic circuits, symbolic representations of such circuits indicating different valves, actuators, etc	4L
7	Module 7: Basics of 8085 microprocessor, programmable register architecture, buses, memory mapping, clock pulse and data transfer operations, and simple assembly and mnemonic programming on 8085 microprocessors.	5L
	Module 8:	

8	Use of On-Off, PI and PID controllers to control different drives, Programming in PLC controller using Ladder diagram.	4L
9	Module 9: Mathematical modeling of physical systems, such as spring-mass vibration system, linear and rotary motion and its Laplace Transform.	2L
10	Module 10: Basics of time domain analysis, Introduction to discrete-time systems and Z-transform.	2L
11	Module 11: Introduction to Mechatronic systems, such as automatic brake, door closing and opening, robot, CNC machine, AGV, etc.	2L
Total		36L

Course Outcomes:

After completion of the course, students will be able to:

1. Model and analyze mechatronic systems for an engineering application
2. Identify sensors, transducers and actuators to monitor and control the behaviour of process or product.
3. Develop PLC programs for an engineering application.
4. Evaluate the performance of mechatronic systems.

Learning Resources:

1	M.D. Singh and J.G. Joshi, Mechatronics, Prentice Hall of India, 2006.
2	D. Shetty and R. Kolk, Mechatronics System Design, 3rd Edition, PWS Publishing, 2009.
3	D.G. Alciatore & M.B. Histan, Introduction to Mechatronics and Measurement systems, 4 th Edition, McGraw Hill, 2006.
4	A. Smaili and F. Arnold, Applied Mechatronics, Oxford University Press, Indian Edition, 2007.
5	K.K. Appu Kuttan, Introduction to Mechatronics, Oxford University Press, New Delhi, 2007.

Open Elective -II

Subject Code	Subject Name	Offered by
OE-EC701C	Mobile Computing	ECE Department
OE-HU701B	Human Resource Development and Organizational Behaviour	MSH Department
OE-IT701A	Introduction to Bioinformatics	IT Department
OE-IT701B	Cyber law and Security Policy	IT Department
OE-ME701D	Robotics	ME Department

Course Name:	Mobile Computing		
Course Code:	OE-EC701C	Category:	Open Elective Course
Semester:		Credit:	3
L-T-P:	3-0-0	Pre-Requisites:	Knowledge of Communication
Full Marks:	100		
Examination Scheme:	Semester Examination: 70	Continuous Assessment: 25	Attendance: 05

Course Objectives:	
1	To learn about different PCS, GSM, GPRS System
2	To learn about overall wireless communication system
3	To learn about different satellites regarding wireless communication

Course Contents:		
Module No.	Description of Topic	Contact Hrs.
1	(PCS): PCS Architecture, Mobility management, Networks signaling. Global System for Mobile Communication (GSM) system overview: GSM Architecture, Mobility management, Network signaling.	5
2	Architecture, GPRS Network Nodes. Mobile Data Communication: WLANs (Wireless LANs) IEEE 802.11 standard, Mobile IP.	5
3	Internet standard, WAP Gateway and Protocols, wireless mark up Languages (WML). Wireless Local Loop(WLL): Introduction to WLL Architecture, wireless Local Loop Technologies.	8
4	International Mobile Telecommunications 2000 (IMT 2000) vision, Wideband Code Division Multiple Access (W-CDMA), and CDMA 2000, Quality of services in 3G.	8
5	IRIDIUM and GLOBALSTAR systems. Wireless Enterprise Networks: Introduction to Virtual Networks, Blue tooth technology, Blue tooth Protocols.	8
6	Server Side programming in JAVA, Pervasive web application architecture, Device independent example application	4
7	Introduction of 4G and 5G Communication	2
Total		40

Course Outcomes:

After completion of the course, students will be able to:

1	Identify the key components of Mobility Management and GSM Architecture
2	Explain Wireless Application Protocol and GPRS Architecture
3	Describe 3G, CDMA, WCDMA
4	Differentiate between IRIDIUM and GLOBALSTAR systems

Learning Resources:

1	Mobile Communication", J. Schiller, Pearson
2	Mobile and Personal Communication systems and services", Raj Pandya, Prentice Hall of India, 2001.
3	Third Generation Mobile Telecommunication systems", by P.Stavronlakis, Springer Publishers
4	Guide to Designing and Implementing wireless LANs", Mark Ciampa, Thomson learning, Vikas Publishing House, 2001
5	Brijesh Gupta "Mobile Computing", Khanna Publishing House, New Delhi
6	Vijay K. Garg, Wireless Communication and Networking

Course Name:	Human Resource Development and Organisational Behaviour		
Course Code:	OE-HU701B	Category:	Open Elective Courses
Semester:	7 th	Credit:	3
L-T-P:	3-0-0	Pre-Requisites:	To know the existence of an organization as a place for human livelihood
Full Marks:	100		
Examination Scheme:	Semester Examination: 70	Continuous Assessment: 25	Attendance: 05

Course Objectives:	
1	To help the students to develop cognizance of the importance of human behavior and how to align it with basic organizational theories.
2	To enable students to describe how people behave under different conditions and understand why people behave as they do.
3	To provide the students to analyze specific strategic human resources demands for future action.
4	To enable students to synthesize related information and evaluate options for the most logical and optimal solution such that they would be able to predict and control human behaviour and improve results.

Course Contents:		
Module No.	Description of Topic	Contact Hrs.
1.	Introduction of Human Resource Development: Human aspect of management, Human Relations; Human Resource Management- Concept, Scope and Importance; Competencies of HR Manager, Human Resource Planning, Job Analysis, and Job Design: Job analysis; Job description and specifications; Job design; Job characteristic approach to job design.	5
2.	Recruitment, Selection, Training, and Development: Factors affecting recruitment; Sources of recruitment (internal and external); Basic selection model; Psychological tests for selection; Interviewing; Placement and Induction; Job Changes- Transfers, Promotions, and Separations; An overview of Training and Development; Emerging trends in Recruitment, Selection, and development.	5
3.	Performance Appraisal, and Audit: Performance Appraisal: Concept, Objectives and Methods; Traditional and Modern Methods- MBO, 360 Degree Appraisal, Behaviourally Anchored Rating Scale, Potential Appraisal, Human Resource Audit.	4
4.	Introduction of Organizational Behavior : Introduction, definition, historical development, An OB model; contributing disciplines, challenges and opportunities. Personality: Meaning, formation, determinants, traits of personality, big five and MBTI, personality attributes influencing OB. Attitude: Formation, components of attitudes, relation between attitude and behavior. Learning; Perception: Process of perception, factors influencing perception, link between perception and individual decision-making; Transactional Analysis: An Introduction to Transactional Analysis; Johari window.	6

5	Group Dynamics and Team Development: Group dynamics -definition and importance, types of groups, group formation, group development, group composition, group performance factors; Principle-centered-approach to team development	4
6.	Motivation: Meaning, theories of motivation-needs theory, two factor theory, Theory X and Y, application of motivational theories. Job satisfaction. Case Study analysis. Leadership: Meaning, styles of leadership, leadership theories, trait theory, behavioral theories, managerial grid, situational theories.	6
7.	Power and Authority: Definition of Power –Types of Power; Power and Politics in Organization; Organizational Stress; Conflict: Nature of Conflict & Conflict Resolution; Case Study Analysis	3
8.	Organizational Change and Development: Planned Change & OB Techniques; Organizational Development; Organizational Culture: Meaning & Definition, Contemporary Models of Culture and Organizational Effectiveness; Cross Cultural Management	3
		36L

Course Outcomes:

After completion of the course, students will be able to:

1	Demonstrate the applicability of the concept of organizational behavior to understand the behavior of people in the organization
2	Demonstrate the applicability of analyzing the complexities associated with management of individual behavior in the organization.
3	Analyze the complexities associated with management of the group behavior in the organization
4	Demonstrate how the organizational behavior can integrate in understanding the motivation (why) behind behavior of people in the organization.

Learning Resources:

1.	D'Cenzo, David A., Stephen P. Robbins, and Susan L. Verhulst, Human Resource Management, John Wiley and Sons, New Delhi
2.	Saiyadain, Mirza S., Human Resource Management, Tata McGraw-Hill Pub. Co. Ltd., New Delhi.
3.	Robbins, S.P. Judge, T.A. & Sanghi, S.: Organizational Behaviour, Pearson
4.	Luthans, Fred: Organizational Behaviour, McGraw Hill
5.	Newstrom J.W. & Devis K.: Organizational Behavior, McGraw Hill
6.	Aswathappa, K : Organisational Behaviour ,Himalaya Publishing House
7.	Shukla, Madhukar : Understanding Organizations – Organizational Theory & Practice in India, Prentice Hall
8.	Sekharan, Uma: Organisational Behaviour , The McGraw –Hill Companies

Course Name:	Introduction to Bioinformatics		
Course Code:	OE-IT701A	Category:	Open Elective Courses
Semester:	7 th	Credit:	3
L-T-P:	3-0-0	Pre-Requisites:	Basic Knowledge of Biology & Database
Full Marks:	100		
Examination Scheme:	Semester Examination: 70	Continuous Assessment: 25	Attendance: 05

Course Objectives:	
1	To exposed to the need for Bioinformatics technologies.
2	To be familiar with the modeling techniques..
3	To exposed to Pattern Matching and Visualization.

Course Contents:		
Module No.	Description of Topic	Contact Hrs.
1	Introduction: Need for Bioinformatics technologies – Overview of Bioinformatics technologies Structural bioinformatics – Data format and processing – Secondary resources and applications – Role of Structural bioinformatics – Biological Data Integration System.	8L
2	Introduction to Molecular Biology: Concepts of Cell, tissue, types of cell, components of cell, organelle. Functions of different organelles. Concepts of DNA: Basic Structure of DNA; Double Helix structure; Watson and crick model. Exons and Introns and Gene Concept. Concepts of RNA : Basic structure, Difference between RNA and DNA. Types of RNA. Concept of Protein: Basic components and structure. Introduction to Central Dogma: Transcription and Translation Introduction to Metabolic Pathways	12L
3	Sequence Databases: Introduction to Bioinformatics. Recent challenges in Bioinformatics. Protein Sequence Databases, DNA sequence databases. sequence database search programs like BLAST and FASTA. NCBI different modules: GenBank; OMIM, Taxonomy browser, PubMed.	12L
4	DNA Mapping and Assembly: Size of Human DNA, Copying DNA: Polymerase Chain Reaction (PCR), Hybridization and Microarrays, Cutting DNA into Fragments, Sequencing Short DNA Molecules, Mapping Long DNA Molecules. DeBruijn Graph. Sequence Alignment: Introduction, local and global alignment, pair wise and multiple alignment, Dynamic Programming Concept. Alignment algorithms: Needleman and Wunsch algorithm, Smith-Waterman.	6L
Total		38L



MCKV INSTITUTE OF ENGINEERING

NAAC Accredited "A" Grade Autonomous Institute under UGC Act 1956
Approved by AICTE & affiliated to Maulana Abul Kalam Azad University of Technology, West Bengal

243 G.T. Road (N), Liluah, Howrah- 711204, West Bengal, India

Ph: +91 33 26549315/17 Fax +91 33 26549318 Web: www.mckvie.edu.in/

Course Outcomes:

After completion of the course, students will be able to:

1	Develop models for biological data
2	Apply pattern matching techniques to bioinformatics data – protein data genomic data
3	Apply microarray technology for genomic expression study

Learning Resources:

1	Bioinformatics Technologies by Yi-Ping Phoebe Chen (Ed). First Indian Reprint, Springer Verlag, 2007
2	Bio Informatics Computing by Bryan Bergeron, Pearson Education
3	Introduction to Bioinformatics by Arthur M Lesk Oxford University Press
4	Bioinformatics for Beginners by Supratim Chaudhury Elsevier.
5	Bioinformatics Algorithms: An Active Learning Approach, Volume 1 by Phillip Campeau Active Learning Publishers
6	Algorithms in Bioinformatics: A Practical Introduction by Wing-Kin Sung CRC Press

Course Name:	Cyber Law and Security Policy		
Course Code:	OE-IT 701B	Category:	Open Elective-II
Semester:	7 th	Credit:	3
L-T-P:	3-0-0	Pre-Requisites:	
Full Marks:	100		
Examination	Semester	Continuous	Attendance: 05
Scheme:	Examination: 70	Assessment: 25	

Course Objectives:

1	To familiarize with the basic terminology related to cybercrime and cyber security.
2	To familiarize with the security challenges faced by mobile devices.
3	To describe the Tools and Methods used in cybercrime.
4	To impart knowledge of cyber forensic evidence gathering and report generation
5	To motivate to analyze cybercrime, its ethical issues and apply different sections of Indian IT Act on it.

Course Contents:

Module No.	Description of Topic	Contact Hrs.
1	1A: Introduction of Cybercrime & Cyber Security: Importance and challenges Cyberspace, Cyber threats, Hacking, Types of cybercrime and cyber criminals.	5
	1B: Steps and categories of Cybercrime: Planning of attacks, social engineering, passive attack, Active attacks, cyber-stalking, Phishing methods, ID Theft and consequences.	5
2	Cybercrime Mobile & Wireless devices: Security challenges posted by mobile devices, cryptographic security for mobile devices, Attacks on mobile/cell phones, Different viruses, Worms, Trojans, Backdoor attacks on laptops and other devices.	5
3	Tools and Methods used in Cyber crime: Proxy servers, password checking, Random checking, Trojan Horses and Backdoors, DOS & DDOS attacks, SQL injection, buffer over flow.	5
4	Cyber Forensics: Introduction to Cyber Forensics, Computer Equipment and associated storage media, Role of forensics Investigator, Forensics Investigation Process and Collecting Network based Evidence. Writing Computer Forensics Reports.	8
5	Cyber Laws: Legal aspects, Indian IT Act, its subsections and case studies, Computer Offences and its penalty under IT Act 2000.	6
	Cyber Ethics, Software piracy, Intellectual Property Rights in Cyberspace.	2
Total		36L

Course Outcomes:

After completion of the course, students will be able to:

1	Recall the basic terminology related to cybercrime and cyber security.
2	Identify the security challenges faced by mobile devices.
3	Describe the tools and methods used in cybercrime.
4	Explain steps of cyber forensic evidence gathering and report generation
5	Use different sections of Indian IT Act on cybercrimes



MCKV INSTITUTE OF ENGINEERING

NAAC Accredited "A" Grade Autonomous Institute under UGC Act 1956
Approved by AICTE & affiliated to Maulana Abul Kalam Azad University of Technology, West Bengal

243 G.T. Road (N), Liluah, Howrah- 711204, West Bengal, India

Ph: +91 33 26549315/17 Fax +91 33 26549318 Web: www.mckvie.edu.in/

Learning Resources:	
1	Cyber Security, Nina Gobole & Sunit Belapune; Pub: Wiley India.
2	Information Security and Cyber Laws, Pankaj Agarwal
3	Information Security & Cyber Laws, Gupta & Gupta, Khanna Publishing House, (AICTE Recommended Textbook- 2018)
4	Computer and Cyber Security ,Principles, Algorithm, Applications, and Perspectives Edited by Brij B. Gupta , Taylor & Francis Group , 2019

Course Name:	Robotics		
Course Code:	OE-ME701D	Category:	Open Elective Courses
Semester:	Seventh	Credit:	3
L-T-P:	3-0-0	Pre-Requisites:	Manufacturing Technology and Basics of automation
Full Marks:	100		
Examination Scheme:	Semester Examination: 70	Continuous Assessment: 25	Attendance: 05

Course Objectives:

1	To impart knowledge about the engineering aspects of Robots and their application
---	---

Course Contents:

Module No.	Description of Topic/ Experiment	Contact Hrs.
1	Module 1: Introduction: Basic concepts- Robot anatomy- Manipulators kinematics: Forward and inverse kinematics- Precision movement, robot specifications and Work volume, Types of Robot drives- Basic robot motions- Point to point control, continuous path contour.	8L
2	Module 2: End Effectors: End effectors- classification- mechanical, magnetic, vacuum and adhesive gripper- gripper force analysis and design. Robot control- Unit control system concept- servo and non-servo control of robot joints, adaptive and optimal control.	7L
3	Module 3: Sensors: Sensor devices, Types of sensors- contact, position and displacement sensors, Force and torque sensors- Proximity and range sensors- acoustic sensors- Robot vision systems- Sensing and digitizing- Image processing and analysis.	6L
4	Module 4: Robot Programming: Robot language classification- programming methods- off and on line programming- Lead through method- Teach pendent method- VAL systems and language, simple program.	8L
5	Module 5: Machine loading and unloading, Assembly, Inspection, Welding, Spray painting, Mobile robot, Micro-robots- Recent developments in robotics- safety consideration.	7L
Total		36P

Course Outcomes:

After completion of the course, students will be able to:

1	Familiarize the basics of robots control system.
2	Demonstrate knowledge of industrial robots, characteristics, end effectors and actuators
3	Apply spatial transformation to obtain forward and inverse kinematics
4	Explain industrial robotics and their applications
5	Explain robot programming

Learning Resources:

1	M.P. Groover. Industrial Robotics Technology Programming and Applications, McGrawHill Book Co, Singapore, 1987.
2	S.K. Saha, Introduction to Robotics, McGraw-Hill Publication, 2014.
3	Y. Koren, Robotics for Engineers, McGraw Hill, New York, 1985.
4	P.G. Ranky and C.Y. Ho, Robots Modelling Control and Applications with Software, Springer Verlag, 1985.
5	J.J. Craig, Introduction to Robotics, Addison-Wesley, 2009.